CLASS 8 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 8: **Nature of Matter: Elements, Compounds and Mixtures** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which of the following is correct for compounds? 1

(a) constituents cannot be separated by simple physical methods
(b) they have no fixed composition
(c) they are elements
(d) they are mixtures

2. Assertion (A): Mercury is used in thermometers. Reason (R): Mercury is a metal that is liquid at room temperature. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

3. A non-metal oxide dissolved in water generally gives a solution that turns: 1

(a) turmeric red **(b)** red litmus blue **(c)** no change **(d)** blue litmus red

4. A mixture of chalk powder and water is shaken and left. The chalk settles. This shows the mixture is: 1

(a) a solution **(b)** an element
(c) a compound **(d)** a suspension (heterogeneous)

5. Assertion (A): Diamond and graphite have different properties although they are both made of carbon. Reason (R): Their atoms are arranged differently. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. Which of the following will be a good thermal insulator? 1

(a) iron **(b)** aluminium **(c)** copper **(d)** wood

7. Which of the following is not a physical method of separation? 1

(a) magnetic separation **(b)** evaporation
(c) filtration **(d)** electrolysis of water

8. Assertion (A): Aluminium foil can be made from aluminium metal. Reason (R): Aluminium is malleable. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

9. Which is a heterogeneous mixture? 1

(a) a salad **(b)** air **(c)** brass **(d)** sugar solution

10. When sugar is heated strongly it turns black and releases steam. This shows sugar is a: 1

(a) compound **(b)** mixture **(c)** metal **(d)** element

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. An element X reacts with oxygen to form an oxide that is acidic. Is X a metal or non-metal? Give an example. 2

12. Write the number and type of atoms in one molecule of glucose, $C_6H_{12}O_6$. 2

13. State two limitations of classifying elements only as metals and non-metals. 2

14. State why an iron nail dipped in copper sulphate becomes brown and the solution fades. Name the type of change. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain why a brass key does not rust but an iron nail does. 3

16. Explain why brass and bronze are not elements and not compounds. 3

17. Explain why the electrical wires are usually made of copper while the outer cover is plastic. 3

OR

List the steps to separate a mixture of iron filings, ammonium chloride and sand with the property used in each. 3

18. Explain why mercury, bromine and gallium are special among elements. 3

OR

Why do we say that air, sea water and soil are mixtures but not pure substances? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain why alloys like stainless steel, brass and solder are preferred in industry, with composition and properties. 5

OR

Write the names, symbols and uses of five metals and five non-metals, and decide which are good conductors. 5

20. Compare physical changes with chemical changes using examples of mixtures and compounds. Include dissolving, heating iron-sulphur, rusting and melting wax. 5

OR

A sample of ammonium chloride and sand is given. Describe the separation and explain the property used. 5

END OF QUESTION PAPER — Total: 40 Marks